

Two regimes of bioinspired memory: an architecture-search perspective

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Abstract—A robot learning across its service life must decide what to keep in fixed-size memory. The episodic buffer in e-MDB, a cognitive architecture for lifelong robot learning, discards the oldest entry whatever its value. We present EMBER (Emergent Memory-Based Encoding and Reactivation), which treats such a memory as a six-axis search space: how a trace is read, written, weighted by surprise, decayed, evicted, and reinforced. Their combinations give 576 configurations of one engine; we evaluate every one, including a nearest-neighbour proxy of the e-MDB buffer. Each is scored on how many important-marked experiences survive a stream overrunning the memory many times over. The memory never sees the marks, only a surprise signal. We report four results. First, which mechanism matters depends on $r = K_{proto}/C$, recurring situations divided by capacity. Below $r = 1$ the write axis explains 73.0 % of the variance in retention; above it, 1.6 %, and eviction takes over. The switch falls at the same r for three capacities, which we state as a hypothesis for the conditions tested, not a general law. Second, the best configuration retains 0.983 of the marked experiences against 0.050 for that proxy, differing on two axes only: surprise-weighted encoding and weakest-trace eviction. Neither alone changes anything. Third, a purely spiking circuit fails the reconstruction test the others pass: it was built for discrete patterns, not continuous vectors. Fourth, two preconditions appear only on real data. Merging needs repeated encounters to look alike; embeddings of natural images do not (0.401 against 0.85). Surprise must come from reward, not perceptual novelty, raising retention from 0.0 % to 52.6 %.

Index Terms—bioinspired memory, neural architecture search, Sparse Distributed Memory, engram consolidation, episodic memory, lifelong learning

I. INTRODUCTION

A cognitive robot that learns over its lifetime faces a memory problem that goes well beyond mere storage. It must decide what is worth keeping, keep that information accessible in proportion to its relevance, and retrieve it reliably even from a partial or degraded cue. These three requirements map onto three well-studied properties of biological memory: selective encoding modulated by surprise and novelty, use-regulated trace consolidation and decay, and content-addressable retrieval that is robust to noise.

The episodic buffer of the e-MDB (*epistemic Multilevel Darwinist Brain*) architecture [1], [2] —in its public ROS2 implementation, a fixed-capacity FIFO (first-in-first-out) queue [3]— satisfies none of the three. It keeps whatever was perceived most recently, treating every experience alike regardless of its importance, novelty or prediction error. Episodes are read back by position or as whole training batches, not by content. When the buffer fills, the oldest episode is simply discarded: it is, in effect, a database with no criterion.

The bioinspired-memory literature offers several principled remedies. Kanerva’s Sparse Distributed Memory (SDM) [4] provides content-addressable retrieval over a distributed superposition of memory locations, with graceful degradation under noise. Hebbian engram consolidation [5], [6] offers a biologically plausible mechanism for reinforcing traces through repeated co-activation. Prediction-error gating regulates how strongly a new experience is encoded as a function of how surprising it is, grounded in dopaminergic modulation of hippocampal encoding by novelty and motivational significance [7], [8] and in the finding that larger reward-prediction errors improve episodic memory for the events that produced them [9]. And forgetting need not be indiscriminate: memory availability tracks how likely an item is to be needed again [10], and transience is an actively regulated process that serves decision-making rather than a failure of storage [11], which motivates eviction by estimated value rather than by age when capacity is exceeded.

But these mechanisms are almost always evaluated in isolation, embedded in architectures purpose-built to show them off, and compared against baselines that do not share their design space. That makes it hard —sometimes impossible— to tell whether a reported gain comes from the mechanism itself or from an incidental advantage of the evaluation protocol.

This work reports on EMBER (Emergent Memory-Based Encoding and Reactivation), an investigation of long-term episodic memory for cognitive robots, and it inverts the usual approach. Rather than proposing a new architecture and arguing that it beats a simple baseline, we define a discrete design space in which each bioinspired mechanism is an axis, and we traverse it exhaustively. The e-MDB FIFO —more precisely, a FIFO-NN proxy of it that adds nearest-neighbour retrieval— is not an external baseline but one more point inside that space, which turns a potentially circular comparison into a falsifiable question: where does the deployed *incumbent* land, and which combination of mechanisms defines the *performance frontier*, the space’s best-scoring point?

We then evaluate the bioinspired architectures that emerge from that analysis with a two-phase benchmark separating the ability to reconstruct a degraded cue —a prerequisite of any associative memory— from retaining the right episodes under continuous capacity pressure. Finally, we instrument the regime hypothesis and the salience mechanism on two real domains —image embeddings from CIFAR-100 [12], a 100-class benchmark of natural images, and the MiniGrid [13] navigation environment under partial observability— exposing preconditions no synthetic design reveals, one of which we

correct.

Three terms recur throughout. A *surprise signal* is a scalar attached to an experience when it is written: either *novelty*, which the memory computes as one minus the cosine similarity to the closest thing it holds, or *prediction error* in $[0, 1]$, from a model of what was expected. The *salience gate* turns that signal into an encoding weight. *Rare-event retention* is the fraction of importance-tagged episodes still recoverable after a stream longer than the capacity (sixteen times longer in the main search); the tag is ground truth used only for scoring, never visible to the memory, which is what keeps the measurement from being circular and why eviction by age is worst placed under it.

Two more terms recur from Section III on. A *configuration* is one complete memory design—a choice on each of the six axes; the 576 configurations are enumerated exhaustively, not sampled or optimized. The ratio $r = K_{proto}/C$ relates K_{proto} , the number of *prototypes* in the stream (distinct recurring kinds of experience, formally the latent clusters its keys are drawn from), to C , the *capacity*: the fixed number of traces the memory holds before a write forces an eviction. Thus $r < 1$ means the recurring experience fits with slots to spare, $r > 1$ that it does not; K_{proto} is known on synthetic streams and estimated on real ones (Section V).

The main contributions of this work are:

- A *regime hypothesis*, supported under the conditions evaluated: the ratio $r = K_{proto}/C$ between recurring experience prototypes and memory capacity determines which family of mechanisms governs performance, with a regime transition that reproduces at three distinct capacities of one stationary synthetic generator.
- A demonstration that in a design space of gated mechanisms, main effects mislead. Modulating encoding strength by novelty and prediction error—the most-cited mechanism in the bioinspired-memory literature—explains barely 1.4% of the variance as a main effect, yet conditioned on the one eviction policy that reads the strength signal it improves rare-event retention by a factor of 14.41. The frontier architecture differs from the deployed incumbent on two axes only, and neither is worth installing without the other.
- A two-phase benchmark that admits architectures by reconstruction before exposing them to capacity pressure, revealing that a pure spiking circuit (leaky integrate-and-fire neurons, LIF, with spike-timing-dependent plasticity, STDP) does not transfer to the continuous-vector domain without an intermediate encoding layer.
- Two preconditions that appear only on real data. Consolidation requires that the merge threshold be reachable in the domain, which on CIFAR-100 it is not: there is no r , nominal or effective, at which a compression regime exists. And the salience mechanism requires surprise to be anchored in the variable that decides what matters—reward—rather than in the perceptual novelty of the state; unlike the first, this one we identify and correct.

II. RELATED WORK

EMBER instantiates four memory substrates, evaluated in Section IV: Sparse Distributed Memory (SDM), the Engram Neural Network (ENN, a construct of this work), a pure spiking circuit, and a hybrid Spiking-SDM. Their roots follow, with those of the search method and the deployment target.

A. Sparse Distributed Memory

Kanerva’s Sparse Distributed Memory [4] models associative memory on the high-dimensional structure of the nervous system. Storing information in one location is fragile; instead patterns are distributed across many overlapping *hard locations* with random addresses, so a noisy query still activates enough of the same ones to reconstruct the original. A write activates the locations within a similarity radius of the queried address; a read sums and thresholds those same locations.

Concretely, in the instantiation used here 512 hard locations are drawn once as random unit vectors and never move. A write adds $s \cdot k$ to the counters of the 25 locations (5%) most aligned with the key k , where s is the encoding strength, so the episode is smeared across twenty-five locations and stored in none; a read sums the counters of the query’s 25 nearest locations and normalizes. A noisy query selects a slightly different active set, but the overlap is large enough to reconstruct the right pattern, so accuracy degrades gradually rather than abruptly—the property that matters for a robot, whose sensors are never clean. Our instantiation departs from the textbook model in two ways: activation is by inner product on a continuous address space rather than a Hamming radius, so the radius self-adapts to local density ([14] extends this to hyperdimensional computing); and it is a hybrid, keeping an explicit trace list alongside the counters and returning the stored trace most similar to the reconstruction, so exact payloads survive.

B. Engrams, Hebbian consolidation and STDP

The *engram*—the physical trace a memory leaves in neural tissue—was coined by Semon [15] and confirmed experimentally in modern neuroscience [5]. Its cellular mechanism is long-term potentiation (LTP): neurons that fire together repeatedly strengthen the synapse between them, the principle Hebb [6] distilled into his dictum. The *Engram Neural Network* (ENN) is the name we give in this work—it is not a named architecture from the prior literature—to the minimal construction that turns this principle into an online memory: a matrix of fast Hebbian weights updated without backpropagation, where each write creates a trace or reinforces an existing one above a cosine threshold, with initial strength proportional to prediction error and exponential decay without reinforcement.

Spike-timing-dependent plasticity (STDP) [16] is the same rule at the level of the individual spike: a synapse strengthens when the presynaptic neuron fires just before the postsynaptic one, and weakens in the opposite order. This groups neurons into assemblies, leaving a structural trace that is precisely an

engram. EMBER’s PoC-3 —the third of the project’s proof-of-concept notebooks— verified this on leaky integrate-and-fire neurons: after repeated presentation, intra-group weights rose well above inter-group weights, and a partial cue recruited the rest of its group without activating the others. Whether that mechanism extends to continuous input vectors is what Section IV tests.

C. Architecture search and prioritized replay

Neural architecture search [17] treats architectural decisions as optimizable variables. Its application to the memory mechanism itself is far less explored: Neural Turing Machines [18] and related memory-augmented architectures [19] fix the memory type and learn how to read and write it. We instead make the memory mechanism the search variable, over a fully enumerable space, so that no result can be attributed to an approximate optimizer.

The closest analogue outside neuroscience is deep reinforcement learning. Prioritized experience replay [20] weights which transition to revisit by temporal-difference error —the same “surprising things count more” principle that motivates prediction-error gating— and Neural Episodic Control [21] indexes a differentiable dictionary by similarity, analogous to SDM’s content-addressable read. In both, the priority and read mechanisms are chosen in advance. This work treats those choices as the search space, and measures in which regime each one matters.

Memory selection under a fixed budget also has a long non-biological history. Cache replacement evicts by recency or frequency of use (LRU, LFU and their refinements) [22], [23]; reservoir sampling keeps a uniform sample of an unbounded stream in fixed space [24], and is a strong baseline for replay buffers in continual learning [25]. Rehearsal methods choose which exemplars to keep by herding towards class means [26], by gradient diversity [27], or, closest to this work, by surprise, reward, distribution matching or state-space coverage for life-long reinforcement learning [28]; replay itself mitigates forgetting in that setting [29]. These methods optimize downstream learning with a fixed selection rule; EMBER instead measures retention across a space of rules. Our *fifo*, *min_utility* (least-read, LFU-like) and random eviction options are the simplest members of these families; the design space does not contain reservoir, LRU or prioritized-sampling rules.

D. The e-MDB architecture and the LOLA problem

e-MDB, developed by the Integrated Group for Engineering Research at the Universidade da Coruña, descends from the Multilevel Darwinist Brain [30] and targets the LOLA problem (*Lifelong Open-ended Learning Autonomy*) [31]: a robot that acquires and relates knowledge autonomously and continuously, in domains unknown at design time and with no predefined task boundaries. Its core is a network-memory long-term memory coupled to a motivational engine of needs and drives from which goals are discovered [32], [1]; knowledge lives in *cognitive nodes* —perceptual classes (P-nodes), goals, world and utility models, contexts (C-nodes) and policies

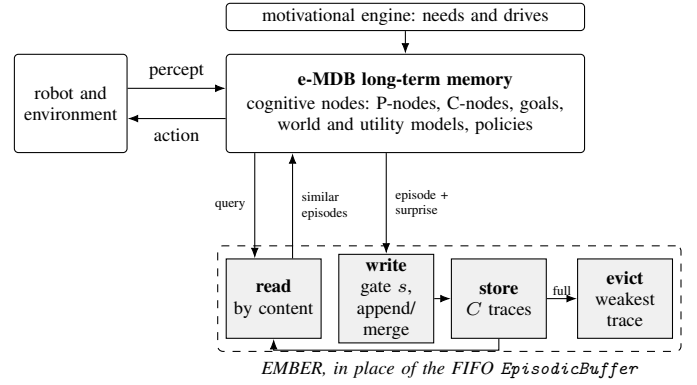


Fig. 1. Where EMBER would sit in e-MDB (schematic; not implemented here). e-MDB activates cognitive nodes in its long-term memory from the current perception and selects actions under its motivational engine [1], [32]; each step yields an Episode that the public implementation appends to a FIFO EpisodicBuffer from which world and utility models are trained [3]. EMBER would replace that buffer: *write* weights each episode by a surprise signal supplied by e-MDB’s own world or utility models, *evict* discards the weakest trace when the store is full, and *read* returns the stored episodes most similar to a query, feeding model training and deliberation.

[2], [3]— which a main cognitive loop activates from the current perception [33]. Its current ROS2 implementation [3] uses an Episode object (*old_perception*, *policy*, *action*, *perception*, *reward_list*) as the unit of experience, and its EpisodicBuffer —the short-term store from which world and utility models are trained— is a bounded deque: episodes are appended, the oldest discarded at the limit, and read back by index or as a (optionally shuffled) batch. There is no importance criterion, no similarity-based retrieval, and no consolidation. The description of the buffer comes from that public code (file *episodic_buffer.py*), not from a publication. In the formulation of Section III its eviction and weighting correspond to the configuration `{read: nn, write: append, strength: constant, decay: 1.0, evict: fifo, reinforce: 0.0}`, one point among 576; because that point retrieves by nearest neighbour, which the real buffer cannot, we call it the *FIFO-NN proxy* of the e-MDB buffer and treat it as an optimistic stand-in, not the deployed code. Fig. 1 shows where EMBER would enter the architecture.

III. SEARCH OVER A BIOINSPIRED DESIGN SPACE

A. The design space

A *configuration* in this work is a complete memory design: one choice on each of six axes, where every axis is a decision an engineer building an episodic memory has to make anyway, usually by intuition. How is a stored experience retrieved —by its single closest match, by a few, or by a whole neighborhood? Is a repeated experience a new memory or a reinforcement of an existing one? Does every experience encode with the same weight, or does surprise weight it? Do unreinforced memories fade? When memory fills, what is discarded? And does recalling a memory strengthen it? Table I lists the six axes, their options, and the biological mechanism each one

TABLE I
BIOINSPIRED DESIGN SPACE. THE CARTESIAN PRODUCT YIELDS 576
CANDIDATE CONFIGURATIONS, EXHAUSTIVELY ENUMERABLE.

Axis	Options	Biological root
read	nn / topk3 / radius	Episodic retrieval vs. activation circle (Kanerva)
write	append / merge	Trace creation vs. Hebbian merge consolidation
strength	constant / novelty / pred_error / both	LTP gating by salience and surprise (dopamine)
decay	1.0 / 0.995 / 0.98	Synaptic weakening without reinforcement
evict	fifo / min_strength / min_utility / random	Forgetting by age vs. by importance
reinforce	0 / 0.5	LTP by reactivation

stands for; Fig. 2 shows how they compose into a single write. Enumerating the Cartesian product —576 configurations— means every combination is tested, including the ones no designer would think to try.

Each configuration is realized by instantiating one parameterized engine with its six values. The 576 are therefore not independently written systems but parameterizations of that single engine, every mechanism implemented exactly once and shared by each configuration that uses it: the same code path runs for every point, so any score difference is attributable to the axis settings alone. Following the NAS literature [17], we reserve *architecture* for Section IV, where it denotes four memory substrates differing in how a trace is physically represented.

The *read* axis operationalizes how past experiences are retrieved: *nn* returns the single closest stored episode; *topk3* aggregates the three closest; *radius* implements the full SDM read, superposing all traces within a similarity radius relative to the best match. The *write* axis controls trace creation: *append* always creates a new trace; *merge* first checks whether a sufficiently similar trace already exists (cosine similarity ≥ 0.85) and, if so, reinforces it instead of creating a redundant copy—the computational analogue of Hebbian consolidation. The *initial-strength* axis—the *strength gate*, which elsewhere in the text we also call the *salience gate* when the emphasis is on which biological signal opens it—modulates how much weight a new experience receives at encoding time: *constant* assigns a fixed weight; the other three options introduce a biological signal—novelty, prediction error, or the two combined. The *decay* axis determines whether the strength of stored traces erodes over time without reinforcement. The *eviction* axis controls what is forgotten when memory fills: *fifo* discards the oldest (the e-MDB default), *min_strength* discards the weakest trace, *min_utility* discards the least-queried, and *random* discards at random. The *read-reinforcement* axis models LTP by reactivation: each time an episode is retrieved, its strength increases.

B. Experimental protocol

Every detail below is taken from the code that produced the results. Keys live in $\mathbb{R}^{32}$ and are normalized to unit length at the memory boundary; every random draw derives from an

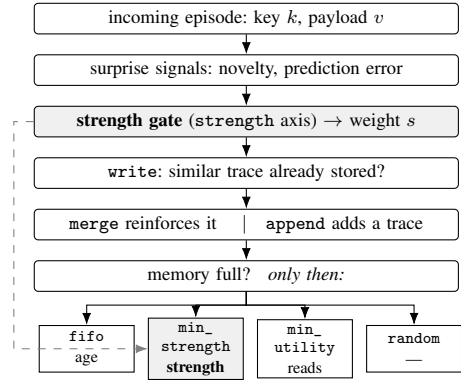


Fig. 2. The write path, and why the salience gate is conditional. The *strength* axis assigns a weight s to every incoming episode, but s is inert unless something downstream reads it. The dashed line marks the only consumer: of the four eviction policies, *min_strength* is the one that consults trace strength, so in three quarters of the design space the salience signal is computed and discarded. Eviction runs only when the memory is already full, which is why the regime transition of Section III falls where it does.

explicit seed, and within a seed all configurations receive the identical stream.

Stream generator. A synthetic stream with K_{proto} prototypes draws K_{proto} centres as i.i.d. Gaussian vectors normalized to the unit sphere (nearly orthogonal at this dimension). Each routine experience picks a prototype uniformly at random, adds isotropic Gaussian noise ($\sigma = 0.05$ per dimension) to its centre and renormalizes; its payload is the prototype identity. Rare experiences have fresh independent random keys and unique payloads, and are inserted at uniformly random positions. The generator also supports correlated centres, power-law visit frequencies and drift; all three are off in every reported experiment. CIFAR-100 streams (Section V) replace centre plus noise by a different image of one of K_{proto} classes per visit, draw rare experiences from disjoint classes, and keep the same prediction-error ranges.

Surprise signals and importance labels. Each experience carries a prediction error drawn from a clipped normal: $\mathcal{N}(0.10, 0.05)$ clipped to $[0, 0.35]$ for routine experiences and $\mathcal{N}(0.90, 0.05)$ clipped to $[0.65, 1]$ for rare ones, so the two ranges do not overlap by construction. Novelty is computed by the memory at write time as one minus the highest cosine similarity to a stored trace (1 on an empty memory). The strength gate assigns $s = 1$ (*constant*), $1 + 2\nu$ (*novelty*), $1 + 2e$ (*pred_error*) or $1 + 2\nu + 2e$ (*both*). The importance label is the generator’s rare flag: it is used only to score and is never passed to the memory, which sees key, payload and prediction error.

Phase 1, reconstruction gate (capacity never binding). R1: 10 patterns, cues keeping 30/50/70 % of the dimensions (the rest zeroed), scored by the cosine similarity of the returned trace when its identity is correct and 0 otherwise. R2: 15 patterns under additive Gaussian noise $\sigma \in \{0.1, 0.2, 0.3, 0.4\}$, scored by identity accuracy. R3: capacity 2, write A then B , read A ; 10 pairs, scored as R1. R4: accuracy at noise $\sigma = 0.15$ for loads $\{2, 5, 10, 15, 20, 30\}$ with capacity equal to load,

averaged. The gate is the mean of R1–R4, with admission at ≥ 0.50 .

Phase 2, battery ($C = 20$). T1, rare-event retention: a stream with $K_{proto} = 5$, 300 routine and 20 rare experiences; every 5 writes the memory is queried with a uniformly chosen, already written key (so read reinforcement can act); at the end each rare experience is queried with its exact key, and counts as retained if its payload is returned at similarity ≥ 0.85 . T2, noisy retrieval under pressure: 60 independent random keys with prediction error $\sim \mathcal{U}(0.05, 0.95)$ are written into 20 slots; 120 queries pick an item with probability proportional to its prediction error and add noise $\sigma = 0.30$; the score is the fraction returning the right payload. T3, sequential interference: 4 blocks of 8 keys, each key written 3 times within its block, prediction error 0.9 in the first block and 0.1 afterwards; the score is the fraction of first-block keys retained as in T1. The *aggregate score* of a configuration is the unweighted mean of T1, T2 and T3, each averaged over seeds.

Seeds and effect sizes. The search and the axis audit use 3 seeds per configuration; the threshold grid, 3 capacities $\times$ 8 ratios $\times$ 5 seeds on T1 alone, with 20 visits per prototype and $\max(5, C/2)$ rare experiences; the substrate benchmark, 4 seeds; CIFAR-100, 5 ratios $\times$ 3 seeds at $C = 20$. For an axis, $\eta^2 = SS_{between}/SS_{total}$ over the 576 configuration scores (the seed-averaged aggregate score in the search; T1 per seed in the grid; seed-averaged T1 on CIFAR-100): a one-way, marginal decomposition that averages over every other axis. The design is a full balanced factorial, so main effects are orthogonal; two-way interactions are computed separately as cell-mean residuals (SS_{ab}/SS_{total}) and are not folded into any axis’s η^2 . Scores are deterministic given the seed, so there is no residual error term and no significance test: η^2 is descriptive. In the grid, η^2 is computed per seed and the reported band is a 95 % percentile bootstrap (2000 resamples) over the five seed-level values —the seed is the bootstrap unit. The crossover of one capacity is the geometric mean of the last write-dominant and first eviction-dominant r ; the interval quoted for the mean crossover bootstraps the three capacities’ crossovers and is only descriptive.

Structural dependence between axes. The axes are not functionally independent, and the factorial decomposition does not assume they are. Trace strength is consumed only by `min_strength` eviction, so the `strength` and `reinforce` axes are inert in the three quarters of the space whose eviction never reads strength; merge acts only when the merge threshold is reached; and under a constant gate without reinforcement `min_strength` reduces exactly to `fifo` (Section III-E). A main effect therefore averages a mechanism over configurations where it cannot act, and a small η^2 is not evidence that the mechanism is weak. We read main effects only as attribution over the whole space, and wherever a dependence is known we also report the conditional effect within the configurations that consume the mechanism’s output. The two-way shares locate the dependence: `strength` $\times$ `evict` explains 4.3 % of the variance, three times the main effect of `strength`,

and `write` $\times$ `evict` 7.8 %.

C. Axis observability

An axis that changes the outcome nowhere is measuring nothing: any conclusion about that mechanism’s irrelevance would be a conclusion about the instrument. Before reporting effects we verify all six axes are observable, by measuring the maximum performance difference between configurations differing only in that axis. All six clear this check, and an axis-liveness audit runs in continuous integration so that none can lapse unnoticed as the space evolves.

Write mode is the strongest main effect (56.6 %), followed by the eviction policy (17.0 %). Decay and initial strength explain much less as main effects (1.8 % and 1.4 % respectively), and read and read-reinforcement come out near zero as main effects (0.0 % and 0.0 %), though both have strictly positive observability (0.119 and 0.206): a null η^2 with positive observability does not say the mechanism does not matter, only that its effect is not a main effect —Section III-E shows exactly that case for initial strength.

The FIFO-NN proxy of the e-MDB buffer —configuration {`read`: `nn`, `write`: `append`, `strength`: `constant`, `decay`: 1.0, `evict`: `fifo`, `reinforce`: 0.0}— scores 0.106, in the range 416–469 of 576, with 54 ties. It is not the absolute floor of the space: below its pessimistic rank of 469 there are configurations that score strictly worse, almost all combining criterion-free append writing with minimum-utility eviction or `topk3` reads over compressed memory. The frontier of the space scores 0.818: a factor of almost eight over the incumbent.

D. The regime transition

The 56.6 % of the write axis is not primarily a property of the merge mechanism; it is a property of the relationship between the number of recurring experience prototypes and memory capacity —the ratio r . We sweep the configuration space over a grid of capacities and ratios, holding visits per prototype fixed so that r is not confounded with stream recurrence.

The transition from “the write axis explains most of the variance” to “the eviction axis does” falls abruptly near $r = 1$ across the three evaluated capacities (10, 20 and 40), independently of K_{proto} : the estimated crossover (geometric mean over the transition cell, averaged over capacities) is 0.88, with a descriptive bootstrap interval over the three capacities of [0.87, 0.89]. The mechanism is simpler than the statistics that localize it. Eviction never runs while the memory has room: a trace is discarded only when a write would exceed capacity. Below $r = 1$ the recurring experience *fits*, merging compresses it into fewer slots, and the freed slots become available for the rare, so what matters is how one writes. Above $r = 1$ there is nothing to compress, memory is saturated, eviction runs on nearly every write, and the question reduces to which policy best preserves what matters. We call these the *compression* and *selection* regimes, and the hypothesis of this section is that which of the two a memory is operating in, rather than

any property of the mechanisms themselves, determines which mechanism matters.

That mechanism also shows why the transition is expected where it falls: it is close to a restatement of what capacity means. When the prototypes fit, consolidation can hold them without displacing anything; when they do not, some form of selection is unavoidable. What the grid adds is that the switch in attribution is abrupt and lands at the same r for three capacities, not that a switch exists. It is established for one stationary generator with equal prototype frequencies, fixed recurrence, isotropic noise, a fixed share of rare events and one merge threshold (0.85); whether r is the governing variable or one component of a broader capacity-pressure relationship under unequal frequencies, drift, noisier similarity, other rare-event prevalences, merge thresholds, stream lengths and a wider capacity range is untested here, and we state it as a hypothesis supported under the evaluated conditions.

Figure 3 plots attribution, not activity: η^2 is how much of the spread in retention across the 576 configurations an axis explains, not how often its operation runs — every configuration in a cell receives the identical stream and performs the identical number of writes. The write and eviction curves are also not the whole decomposition: above the threshold they jointly account for about a fifth of the variance, the rest belonging to the strength gate, decay and their interaction with eviction, which Section III-E isolates.

E. The interaction between salience and eviction

The 1.4% main effect of initial strength is misleading read on its own. Strength modulation influences behavior only when a mechanism *reads* that signal, and of the four eviction policies only `min_strength` does. Conditioning on `{evict: min_strength, write: append}`, a constant gate leaves the policy nothing to rank by (3.6%), whereas novelty and prediction error together make it selective (52.0%): a conditional improvement of $14.41\times$, entirely attributable to this interaction. The factor should be read for what it is. On this benchmark the prediction errors of routine and rare experiences come from non-overlapping ranges (Section III-B), so once an eviction rule reads strength, keeping the rare events follows almost directly from the construction: the number shows that the salience signal has no effect until something consumes it, not that selecting what matters from a realistic signal is solved. How much of the gain survives overlapping, noisy or misleading signals is a separate question.

The same interaction can be shown without statistics, on four concrete configurations. The frontier differs from the incumbent proxy on exactly two of the six axes: the strength gate (`constant` $\rightarrow$ `both`) and the eviction policy (`fifo` $\rightarrow$ `min_strength`). Changing either alone leaves the architecture *bit-identical* to the incumbent, in every combination of the remaining axes (Table II). The reason is exact rather than statistical: under a constant gate with no read reinforcement every trace carries strength 1.0, so selecting the minimum always returns the oldest surviving trace — `min_strength` does not approximate FIFO, it *is* FIFO. Symmetrically, a salience gate

TABLE II
NEITHER MECHANISM WORKS ALONE. MEANS OVER THE THREE BATTERY TASKS.

Configuration	mean	T1	T2	T3
FIFO-NN proxy (incumbent)	0.106	0.050	0.269	0.000
+ salience gate only	0.106	0.050	0.269	0.000
+ <code>min_strength</code> only	0.106	0.050	0.269	0.000
+ both (frontier)	0.818	0.983	0.469	1.000

feeding a policy that never reads strength computes a weight nothing consumes.

F. The frontier architecture

The best-scoring point of the space is `{read: nn, write: append, strength: both, decay: 1.0, evict: min_strength, reinforce: 0.0}`, at 0.818 against 0.106. In practice: every experience is stored as its own trace, written with a weight proportional to how surprising it was. Weights never decay and are never revised on recall, so importance is fixed at encoding. Recall returns the closest trace, and when memory fills the lowest-weight trace is discarded — by construction the least surprising thing held.

Two properties matter more than the rank. It *appends*: the axis that dominates the variance below $r = 1$ is the write mode, but the winning setting never consolidates. Merging compresses routine into fewer slots, yet accumulates strength geometrically on whatever recurs, so a prototype visited thirty times outweighs a rare event encoded once under maximum surprise. And it is *regime-neutral*: never depending on merging, it does not care whether r is above or below the threshold, and always consulting the importance signal it behaves identically in both. That is what makes it a deployable recommendation rather than the winner of one cell — a robot running this configuration need not know, or estimate, its own r .

IV. COMPARATIVE ARCHITECTURE EVALUATION

The search of Section III ranks configurations of one engine on retention under capacity pressure, and the winning one reads by nearest neighbour — a mode that never exercises a distributed representation. It leaves a prior question untested: whether each memory *substrate*, the physical scheme by which a trace is stored, can perform the operation any associative memory must support — reconstructing a complete pattern from a noisy or partial cue. This section runs a two-phase benchmark that first admits substrates by reconstruction fidelity (R1–R4, no capacity pressure) and then re-tests them under continuous pressure (T1–T3). The four are SDM, ENN (pure Hebbian memory), Spiking-SDM (SDM with LIF-neuron activation) and a pure spiking circuit (LIF + STDP), with the FIFO-NN proxy (“FIFO” in Table III) again included as a point in the same space. Table III summarizes both phases: R1 recovers a masked pattern, R2 a noisy one, R3 checks whether B overwrites A , R4 traces accuracy against load; T1

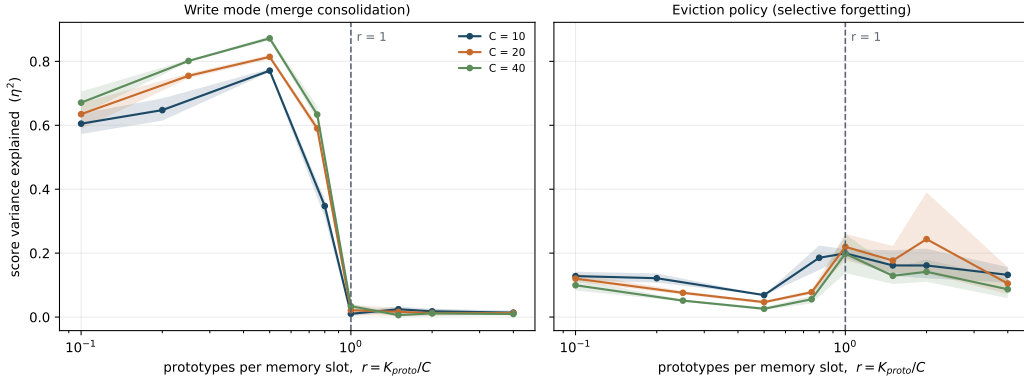


Fig. 3. Where the regime transition falls. Share of the variance in rare-event retention, across the 576 configurations, attributable to the write axis and to the eviction axis, as a function of $r = K_{proto}/C$. This is a test that could have failed rather than a performance plot: if absolute capacity governed the transition the three curves would cross at three different points, and if stream redundancy governed it, holding visits per prototype fixed would have flattened the effect. Neither happens. Note that η^2 is direction-blind — it identifies which axis moves the outcome, not which setting is preferable: the frontier architecture uses append, not merge. Bands: 95 % percentile bootstrap over the five seed-level η^2 values.

TABLE III

TWO-PHASE EVALUATION. R1–R4 MEASURE RECONSTRUCTION WITH NO CAPACITY PRESSURE AND DECIDE ADMISSION; T1–T3 MEASURE RETENTION UNDER PRESSURE.

Arch.	R1	R2	R3	R4	Mean	Gate	T1	T2	T3
SDM	0.672	0.788	1.000	0.978	0.859	✓	1.000	0.365	0.875
ENN	0.691	0.938	1.000	1.000	0.907	✓	0.750	0.344	0.375
Spiking	0.190	0.146	1.000	0.499	0.459	×	—	—	—
SpikingSDM	0.507	0.433	0.950	0.588	0.619	✓	0.675	0.158	0.719
FIFO	0.691	0.938	1.000	1.000	0.907	✓	0.038	0.254	0.000

is rare-event retention, T2 noisy retrieval under pressure, T3 how much of an early block survives later ones.

Capacity here counts traces, which is not the same budget for every substrate: SDM also keeps 512 hard locations with distributed counters next to its explicit trace list, so equal C does not mean equal bytes or equal read/write cost, and the benchmark does not separate the contribution of the counters from that of the trace list.

Every architecture except the pure spiking circuit clears the admission threshold of 0.50 (0.459 against a gate of 0.50). ENN and FIFO get exactly the same gate score: with no capacity pressure, ENN’s merge threshold (0.85) is rarely exceeded between near-orthogonal vectors, so every write creates a new trace and both architectures reduce to nearest-neighbor search over the full set of stored keys. The difference between them emerges only under capacity pressure with structured streams, in phase 2.

The pure spiking circuit fails pattern reconstruction and noise robustness despite forming distinguishable assemblies (perfect $A \rightarrow B$ interference). The cause is a domain mismatch: the circuit was designed and validated on discrete neural activation patterns; over a continuous float32 vector, inner-product neuron selection is sensitive to small perturbations, and a noisy version of a stored vector activates a partially different set of neurons. This result is not a flaw of the spiking mechanism itself — it is the direct motivation for EMBER’s PoC-5: an integrated core combining SDM (for the continuous-to-distributed mapping) with ENN (for the trace

lifecycle).

Phase 2 has a built-in ceiling worth naming. The stream carries twenty rare items into a memory of capacity twenty, with the prediction error of common and rare experiences drawn from non-overlapping ranges, so an architecture that gates on prediction error and evicts by minimum strength should end holding exactly the rare set: high retention confirms correct wiring, not a hard problem. Retrieval is what the score tests independently — each trace must still be recovered from the distributed superposition at similarity ≥ 0.85 — so the informative comparison is SDM against ENN, which shares the gate and eviction policy, not SDM against the FIFO. The threshold grid of Section III puts the rare count at half the capacity, which removes the exact-fit ceiling but not the separability: its prediction-error ranges are the same, so it measures which axis carries the variance, not how hard selection is.

In phase 2, SDM achieves perfect rare-event retention (1.000, $\sigma = 0.000$ across the four seeds): rare events receive high initial strength from prediction error, and minimum-strength eviction always displaces the routine first. ENN, with the same strength gate and the same eviction policy, retains less (0.750, also with no seed-to-seed spread): its merge mechanism, reinforced repeatedly by routine visits, can accumulate more strength than a rare event written only once. Spiking-SDM retains 0.675, and the FIFO — with no importance criterion at all — is the floor among architectures (0.038, $\sigma = 0.041$; the floor holds across seeds). With

a block-differentiated prediction-error signal, the sequential-interference task gives 0.875 for SDM, 0.375 for ENN, and 0.000 for the FIFO—all three with no seed-to-seed spread. This is the comparison that separates the three architectures by their ability to resist catastrophic forgetting.

V. VALIDATION ON REAL PERCEPTUAL EMBEDDINGS

The preceding experiments run on random Gaussian vectors in $\mathbb{R}^{32}$, nearly orthogonal at that dimension. This reduces cross-interference and could overestimate retrieval accuracy relative to real embeddings, which have correlated structure and uneven cluster density. We therefore instrument the same grid on streams built from CIFAR-100 [12] embeddings extracted with a pretrained ResNet-18 [34] and reduced by principal component analysis (PCA) to the working dimension.

The result is not that the threshold shifts: it is that, with this extractor, the write axis is inoperative in all five cells of the sweep, explaining between 0.0 % and 0.4 % of the variance regardless of the ratio. The deciding quantity is *intra-prototype similarity*: how alike two encounters with the same kind of thing look to the memory, as a cosine. Consolidation is possible only when it clears the merge threshold, since merging is triggered by exactly that comparison—if two photographs of one class do not look alike enough, the memory cannot recognize them as the same situation and stores both. The measured median is 0.401, well below the merge threshold (0.85), so merging almost never triggers and $K_{effective}$ —the traces the routine occupies after consolidation—grows without bound instead of converging to K_{proto} : from 84 to 1435 traces as nominal K rises from 5 to 80, with capacity fixed at 20. The synthetic domain in the same run crosses normally ($r \approx 0.77$), so the null result is specific to this extractor, not a failure of the grid.

Intra-prototype similarity is not a term in r ; what it decides is whether r is well defined at all. Where consolidation is reachable but only partial, it acts on the ratio through $K_{effective}$: sub-threshold visits fragment a prototype into several traces, so the effective ratio exceeds the nominal one and the hypothesis holds when stated over it. Where consolidation is unreachable $K_{effective}$ has no stable value—it grows with stream length instead of converging—so this precondition is not a special case of an effective ratio but a separate binary check that comes first, and on CIFAR-100 it already fails: no recomputation of r , nominal or effective, produces a compression regime where there is none.

On a real stream K_{proto} must be estimated. We cluster the keys over a sweep of candidate counts and take the count maximizing the mean *silhouette coefficient*, which for each vector compares its distance to its own cluster against its distance to the nearest other one: near 1 means tight, well-separated clusters, near 0 means the partition is arbitrary. Clustering returns groups whether or not any exist—on isotropic noise the whole sweep scores 0.04–0.07 with no peak, yet still yields an argmax—so we require a silhouette of at least 0.15 before treating r as defined, and report a bootstrap interval rather than a point integer.

On CIFAR-100 that estimation becomes uninformative: the intervals reach the ceiling of the search in several cells, consistent with the absence of compact cluster structure in this domain. This is a further reason to treat merge reachability—measured directly as a similarity comparison, with no clustering involved—as the robust diagnostic, rather than a derived $\hat{K}$.

VI. VALIDATION IN A REAL ENVIRONMENT UNDER PARTIAL OBSERVABILITY

CIFAR-100 tests the regime hypothesis on real data, but it is still a hand-built stream from static embeddings. A robot does not receive a stream: it receives the consequences of its own actions, with partial observation and temporal structure that no generator reproduces. We instrument the `ember.envs` adapter on MiniGrid-MemoryS13-v0 [13], a navigation environment designed specifically to demand memory: the agent must remember an object seen at the start of the episode in order to choose correctly at the end. We compare the frontier configuration from Section III against the FIFO with `t1_rare_retention`—the same task that measures rarity throughout the rest of the program—across three conditions that change one variable at a time (Table IV).

Baseline. With a uniform random policy, both the frontier and the FIFO sit at the rare-event retention floor. The cause is not a retrieval error—with no capacity pressure the frontier recovers any stored trace with similarity 1.0—but that the one-step prediction error, computed by an online linear transition model over the observation, does not separate the rewarded from the merely novel: 75.4 % of common experiences have a prediction error equal to or greater than the least surprising rewarded event, measured over 38 rewarded events. This quantity is the *overlap* reported in Table IV: the smaller it is, the more cleanly the surprise signal separates what mattered from what did not.

Ruling out data scarcity. The simplest explanation must be ruled out first. A policy biased toward moving forward over turning—which does not touch the surprise computation, only what gets visited—raises the rewarded-episode count from 38 to 156 on the same step budget. Retention stays at the floor, and the overlap does not improve (73.5 %): quadrupling the data without touching the signal changes nothing, so the problem is not a small sample.

The signal, not the quantity. The strength gate is motivated by the dopaminergic account of prediction error, but in its canonical form that account is about *reward*-prediction error [35], and so is the behavioural evidence that prediction error enhances episodic memory [9]—not perceptual prediction error. We replace the transition model with a reward predictor—same online linear architecture, same random policy as the baseline, changing only what is predicted—and the overlap falls to 5.6 %. The frontier’s retention rises from 0.0 % to 52.6 %—with 38 rare events per condition, a 95 % Wilson interval of [37.3 %, 67.5 %]—while the FIFO—which does not read the strength signal at all—stays at 2.6 % ([0.5 %, 13.5 %]), just as in the baseline. The two intervals do not

TABLE IV
MINIGRID-MEMORYS13-v0: THREE CONDITIONS, 40 ROLLOUTS EACH.

Policy	Surprise	Rare	Overlap	Frontier	FIFO
Random	Perception	38	75.4 %	0.0 %	2.6 %
Forward-biased	Perception	156	73.5 %	0.0 %	1.3 %
Random	Reward	38	5.6 %	52.6 %	2.6 %

overlap despite the small sample, but they treat each rewarded event as an independent trial, whereas events from the same rollout may be correlated; the rollout is the more defensible unit. artifact of the predictor change: it is the salience gate working, now given a signal correlated with what matters.

The salience mechanism does transfer to a real environment under partial observability —not automatically, but under an identifiable and correctable precondition: surprise must be anchored in the variable that decides what matters, which in an environment with reward is reward itself.

VII. DISCUSSION

A. The regime hypothesis and its practical implication

The ratio $r = K_{proto}/C$ has a direct operational reading. In an early, structured environment —a warehouse, a lab— the robot sees the same things repeatedly, $r < 1$, and consolidation is efficient; in a heterogeneous one r grows above 1 and selective eviction by relevance dominates. The recommendation is not to pick a regime: the frontier architecture works in both, never depending on merging (which fails above $r = 1$) and always exploiting the importance signal.

B. The spiking circuit and the motivation for PoC-5

The spiking circuit’s failure at the reconstruction gate is a property of the domain, not the mechanism: PoC-3 formed reliable engrams with LIF+STDP in its native discrete setting. What is missing is the bridge from continuous vectors to discrete assemblies —EMBER’s PoC-5, pairing SDM for the continuous-to-distributed mapping with ENN for the trace lifecycle, as a single *Cognitive Node*.

C. The FIFO contrast, and the precondition as a diagnostic

Mapping the e-MDB FIFO to a configuration is deliberately generous, so the FIFO-NN proxy is an optimistic stand-in and not the deployed buffer: the real *EpisodicBuffer* has no content-addressable retrieval at all, and returns episodes by position or as training batches. We chose nearest-neighbour reads to isolate eviction and weighting, the two axes on which the frontier differs. Even so the proxy ranks 416–469 of 576 and is the floor at retaining rare events among substrates, though not at reconstruction. This is not a criticism of e-MDB, built for a different primary goal —its buffer feeds model training, not recall— but a specific claim: as a store of what to remember, the buffer as implemented is inadequate for the LOLA regime, with a quantified gap for its most favourable proxy.

The finding of Section V gives the robot a check it can run online: intra-prototype similarity, measured on the vectors the memory already receives and compared against the merge

threshold, decides —before the write axis matters at all— whether consolidation is worth attempting, with no external instrumentation.

D. The source of surprise matters as much as its existence

Section VI generalizes this beyond merging. The synthetic and CIFAR-100 designs share a silent assumption: that the surprise signal arrives aligned with what the robot needs to remember. That is free when a generator sets it by hand; in a real environment one must choose *what to predict*, and that choice decides whether the mechanism has anything to protect. The configuration does not change —only the signal it is fed— so the salience gate inherits the quality of its input, and choosing that input is a design decision as consequential as any axis of the space.

VIII. LIMITATIONS AND FUTURE WORK

The real-data validation of Section V uses one extractor (ImageNet-pretrained ResNet-18, PCA to 32 dimensions), one dataset and one capacity; whether the intra-class similarity of 0.401 is a property of the visual domain, the extractor, or both is untested, and a fine-tuned extractor or a per-domain merge threshold could change it. The MiniGrid validation likewise uses one environment and one exploration policy besides the forward-biased one.

Three further limits are not addressed by any experiment in this revision. EMBER has not been integrated into the running e-MDB: Fig. 1 is a design, the incumbent is a proxy rather than the deployed buffer, and whether better retention improves the world and utility models e-MDB trains from its buffer is untested. All real-data evidence comes from a single visual extractor. And no experiment ran on a physical robot.

The most consequential open question is a tension between our strongest result and the setting that motivates the work. Section VI resolves the salience precondition by anchoring surprise in reward —but LOLA is defined by the *absence* of predefined goals, and an open-ended robot may face long stretches with no reward to predict. The available signal then falls back to perceptual novelty, which is precisely where we measured retention at the floor. The fix is valid where a reward exists and unproven where it does not, which is the regime e-MDB is ultimately built for. Intrinsic motivation, learning progress, or the needs and drives of e-MDB’s motivational engine [32] are testable substitutes within the same protocol; none is evaluated here. How to define salience when importance is delayed, ambiguous or unrelated to any external reward remains open, and it is the question on which the usefulness of the salience gate for LOLA ultimately rests. Finally, because no result was obtained on a physical

robot, real sensor noise, continuous control and lifelong non-stationarity remain untested, and the generator’s distributional-drift parameter is never activated in the reported experiments.

IX. CONCLUSION

We present EMBER, an investigation of bioinspired episodic memory for cognitive robots. On the synthetic streams evaluated, exhaustive architecture search indicates that which memory mechanism matters is not fixed but set by the operating regime $r = K_{proto}/C$: below a threshold near 1 the write axis governs retention, above it eviction does—a hypothesis tied to how capacity is defined, whose robustness beyond this generator remains to be shown. The encoding-salience mechanism the literature most motivates has a negligible main effect, yet conditioned on the one eviction policy that reads its signal it recovers fourteen times the rare-event retention—not weak, but gated. The benchmark adds that SDM with a prediction-error gate is the only substrate pairing reconstruction fidelity with perfect retention on a stream whose rare events are separable by construction, that the FIFO-NN proxy of the e-MDB buffer reconstructs well but is the floor at retention—its deficiency is choosing, not remembering—and that the pure spiking circuit does not transfer to continuous vectors, marking PoC-5 as the next milestone. Real data adds two preconditions: consolidation needs intra-prototype similarity above the merge threshold, which CIFAR-100 never reaches, turning a condition of the regime hypothesis into an online-measurable check; and salience needs surprise anchored in reward, under which retention rises from the floor to more than half with no change of configuration or policy. What the salience mechanism needs is the right signal.

CODE AND DATA AVAILABILITY

The code and configuration files for the design space, evaluation tasks and all experiments, together with the measured results (results/**/* .json, with a provenance manifest per experiment) backing every number in this paper, are available to the reviewers through an anonymized repository, and will be released under an open license after review.

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